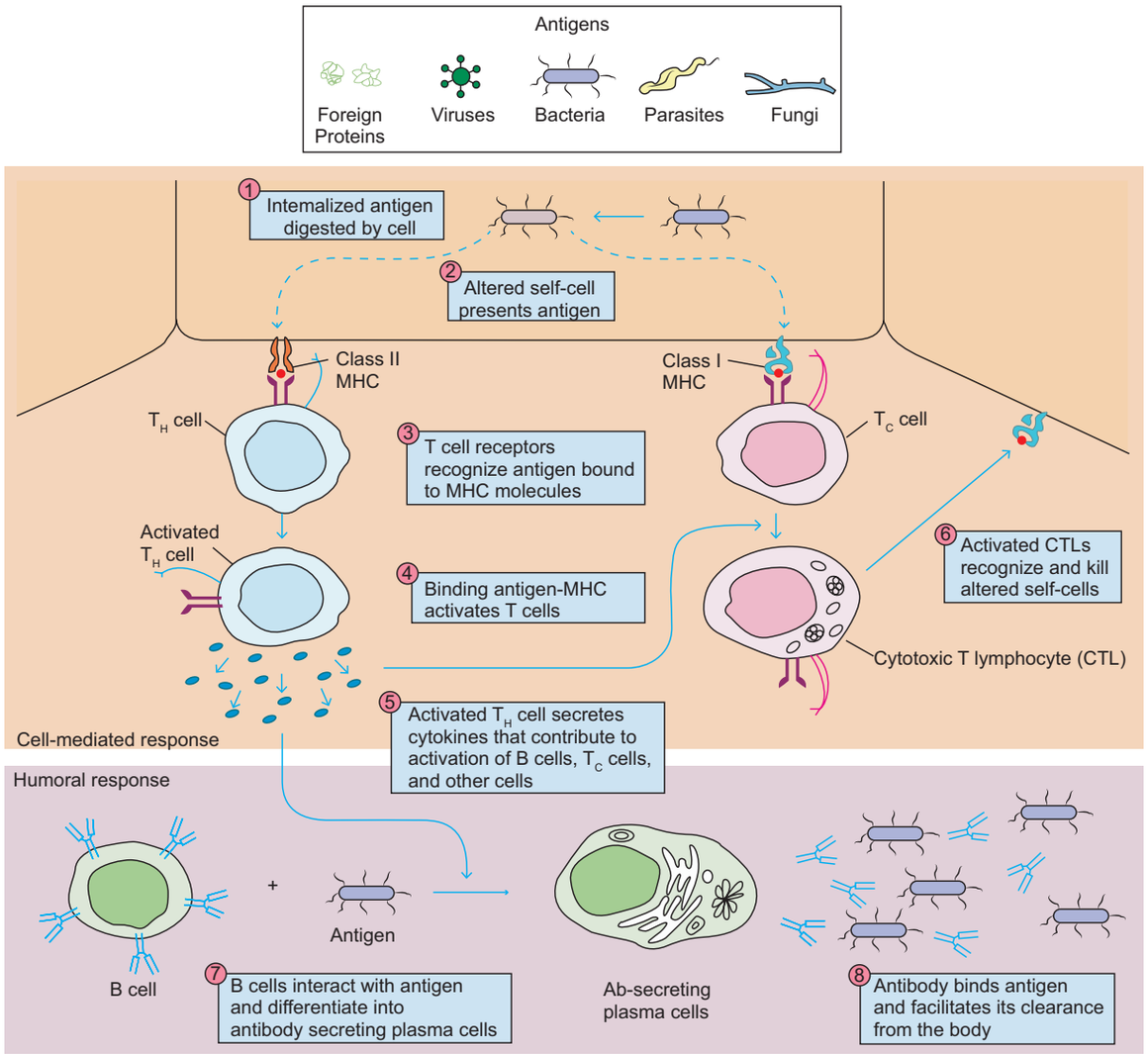


Figure 13.21: Overview of the humoral and cell-mediated branches of the immune system. In the humoral response, B cells interact with antigen and then differentiate into antibody-secreting plasma cells. The secreted antibody binds to the antigen and facilitates its clearance from the body. In the cell-mediated response, various subpopulations of T cells recognize antigen presented on self-cells. T_H cells respond to antigen by producing cytokines. T_C cells respond to antigen by developing into cytotoxic T lymphocytes (CTLs), which mediate killing of altered self-cells (Example: virus-infected cells).



Passive Immuno-therapy: Treatment of an infectious disease by administration of previously generated antibodies specific for the infectious pathogen.

antibody, the binding of the antigen to the antibody causes the cell to divide rapidly. Its progeny differentiate into memory B cells and antibody secreting plasma cells. A single plasma cell can secrete more than 2000 molecules of antibody per second. Circulating antibodies bind to

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